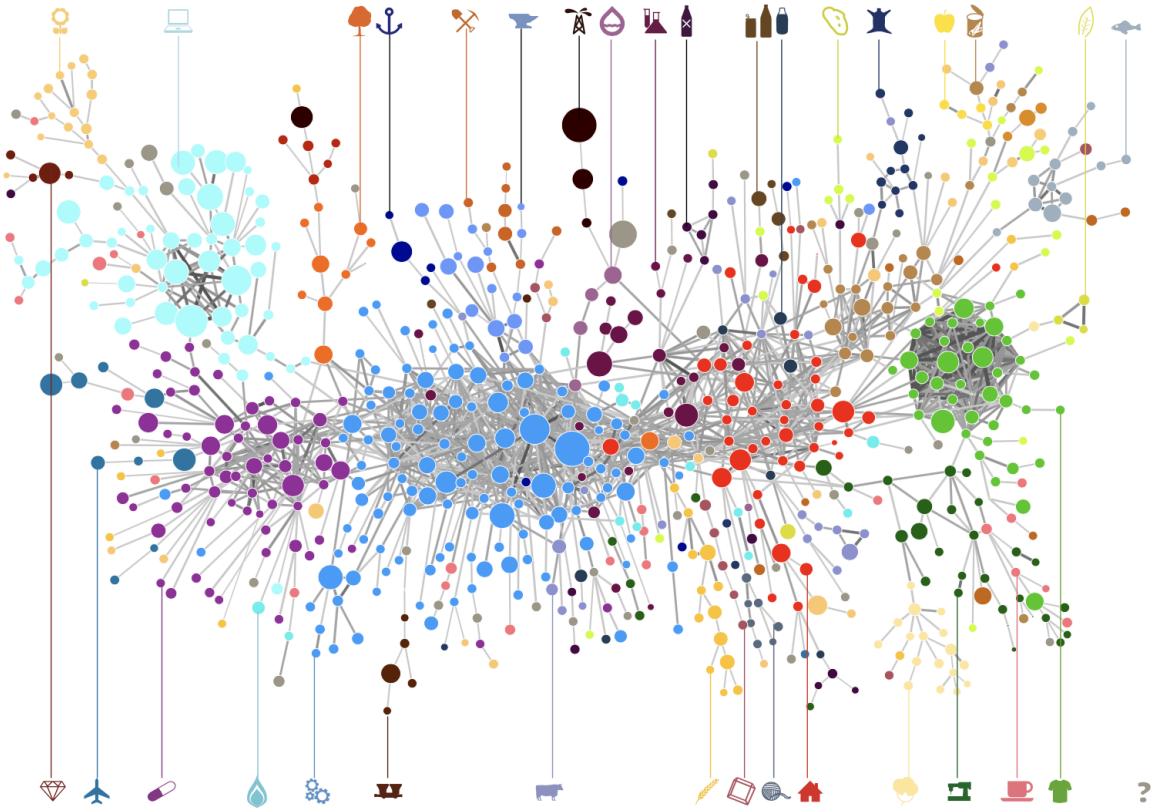


https://atlas.cid.harvard.edu/

Optimal Industry Development Strategy Using Economic and Product Complexity

Singapore

—

The Economic Complexity Lab - Columbia University

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Optimal Industry Development Strategy for Singapore

0. Executive Summary

In this report, we analyze the economic trajectory of Singapore over the period from 2019 to 2024, examining its production capability and structural evolution through the lenses of Gross Domestic Product (GDP) and Economic Complexity Index (ECI). We find that Singapore demonstrated a high level of economic growth, with an average annual GDP growth rate of 7.0% during this period, placing it above its developed peers.

Growth reflected continued depth in electronics, pharmaceuticals, and refined petroleum exports alongside post-pandemic recovery in financial and business services, with FDI weighted heavily toward semiconductor capacity expansion and biopharma manufacturing.

On the complexity front, Singapore's ECI increased over the analysis period, reflecting a standing below the peer average among the comparable developed economies examined. The ECI gain trailed the peer average despite faster GDP growth, suggesting Singapore's expansion in this window leaned on intensifying existing comparative advantages rather than broadening productive diversification.

Examining realized industry development between 2019 and 2024, we find that Singapore progressed across the majority of its industry groups: 18 groups moved closer to global competitiveness thresholds while 0 retreated, a increase from 0 improving groups at the start of the period. Based on our opportunity gain and sensitivity analysis, the industries offering the greatest complexity upside for Singapore are Weapons, Chemical Manufacturing, and Textile. Sensitivity analysis further identifies Transportation as the sector whose marginal expansion yields the largest positive shift in ECI.

Our Structural Optimality Index (SOI) analysis measures the degree to which Singapore's realized investment trajectory aligns with the theoretically optimal complexity-based development path. Singapore ranks in the Q3 quartile globally, with an adjusted R^2 of 0.0036, indicating a medium level of structural alignment. In concrete terms, Singapore has moderately channeled resources toward Animals and By-products, Vegetables, Fruits and Plants, Food and Preparations, Energy Drilling and Mining, Chemical Manufacturing, Raw Manufacturing, Wood Products, Metals, Weapons — sectors where the complexity opportunity is more limited — while underinvesting in None, which offer greater complexity upside. Given the positive correlation between SOI and long-run GDP growth established across our global sample, Singapore's moderate structural alignment is consistent with its high growth record, and suggests that measurable gains remain available through a more complexity-guided development strategy.

1. Report Overview

In this report, we analyze the economic trajectory of Singapore during the period from 2019 to 2024. The window spans the pandemic-era contraction and recovery, the supply-chain reordering of 2021–2023, and intensifying regional competition in semiconductor and biopharma capacity. Specifically, we examine Singapore's macroeconomic trajectory through the lens of Gross Domestic Product (GDP) and Economic Complexity Index (ECI), which reflect a country's production capability and productive structure, respectively. We then analyze the industrial framework and structural changes of Singapore based on quantitative measures including the Product Complexity Index (PCI), followed by a detailed analysis of how those changes aligned with optimal development strategies as measured by opportunity gains at the industry level. We conclude with macro-level results derived from regression models established on a global scale, which take into account countries' positions in the product space.

2. Macroeconomic Results

2.1 GDP Ranking and Changes

Figure 1 illustrates the GDP time series for Singapore and its peer economies (South Korea, Japan, and Germany) from 2019 to 2024. Singapore demonstrated a high average annual GDP growth rate of 7.0% over this period, placing it above peers among the compared economies. For reference, the average annual growth rates of the peer economies were: South Korea at 1.3%, Japan at -3.8%, and Germany at 3.0%.

Japan's USD-denominated GDP declined steadily across the window while Germany's contracted from 2021 onward, narrowing the gap with South Korea. Singapore and the Netherlands stayed on roughly parallel ascending paths from lower bases.

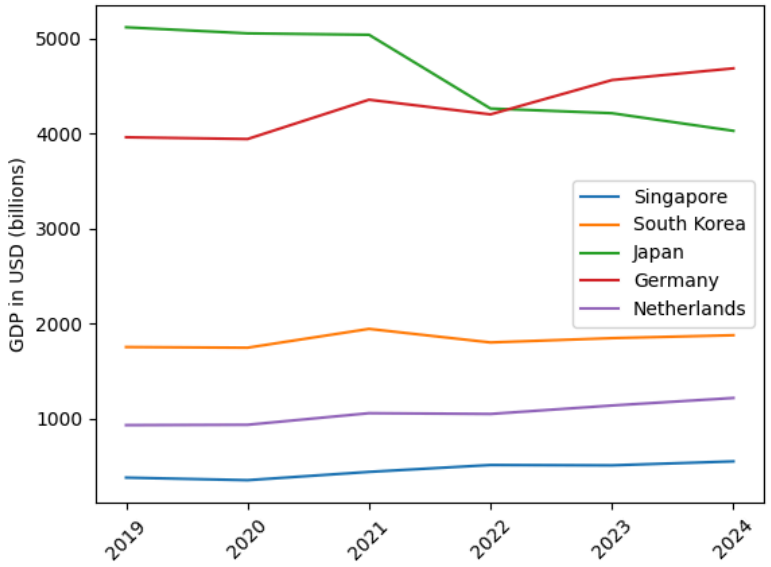


Figure 1. GDP Comparison Among Peer Countries (2019-2024)

2.2 ECI Ranking and Changes

The Economic Complexity Index (ECI), introduced by Hausmann and Hidalgo (2009) to quantify the knowledge embedded in a country's productive structure, is illustrated in Figure 2.

Singapore's ECI increased from -1.276 in 2019 to 2.417 in 2024, a change of +3.692 — a significant shift in productive complexity. Relative to its developed peers, Singapore's ECI trajectory was below the peer average. For context, ECI changes over the same period were: South Korea at +4.263, Japan at +6.180, and Germany at +4.144.

The gain reflected sustained presence in High-complexity Products and Machinery & Construction rather than entry into new categories — consistent with intensification, not diversification.

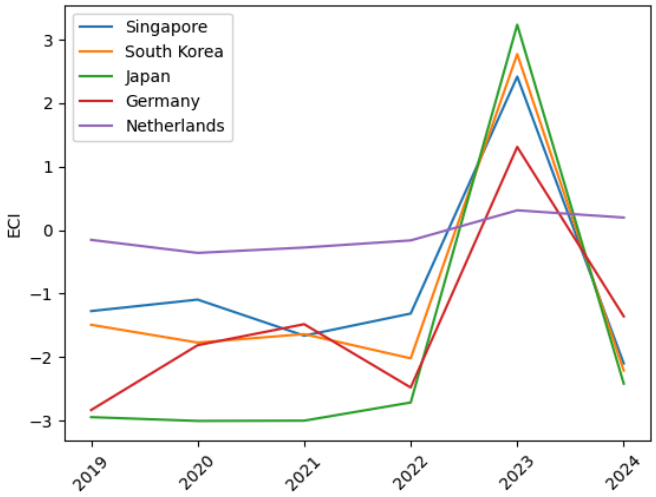


Figure 2. ECI Comparison Among Peer Countries (2019-2024)

Figure 3 provides a combined view of GDP and ECI for Singapore across the full analysis window. The two indicators were partially consistent during this period. ECI moved approximately in tandem with GDP through 2022, then diverged sharply in 2023–2024 — a pattern more consistent with measurement noise in the ECI series than a real structural break.

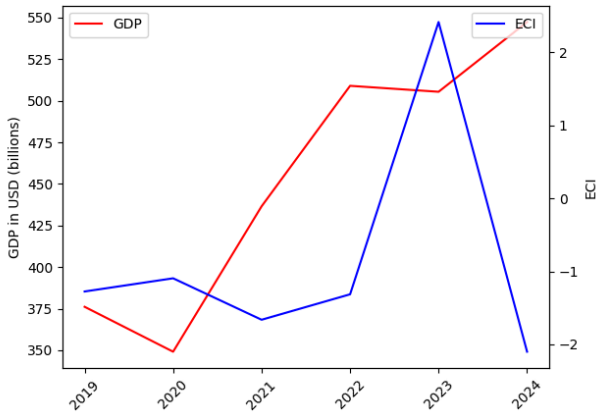


Figure 3. GDP Growth and ECI for Singapore (2019-2024)

3. Production Analysis

3.1 Product Categorization

Our main data source in this research is the UN Comtrade Website (https://comtrade.un.org), which provides us with valuable information on countries' export values on an annual basis, including highly specific export data for more than 5000 categories of products. In order to have a holistic vision of the product space of a country, however, a level of categorization is required. Products in our data source are labeled from 010000-999999, with the first two digits representing a general category to which the product belongs. Based on the 99

categories extracted from the label, we further group the products into the 18 most widely-accepted industries, as summarized in Table 1.

Note that the High-complexity Products industry group is not a distinct industry but rather an aggregated industry that encompasses the most complicated products from the most technologically-advanced industries (as measured by PCI). This industry serves as an indicator for us to quantitatively evaluate how well a country has performed in developing cutting-edge technologies and closing its technological gap with advanced economies, which has become a topic of wide concern today. Indeed, when we rank the 18 categories based on PfCI (an equivalent concept of PCI as calculated based on the 18 categorized product groups), the High-complexity Products industry has the highest PfCI among all industry groups.

Category	Description
Animals and By-products	All kinds of meat, livestock and related by-products
Vegetables, Fruits and Plants	All kinds of plants, edible vegetables and fruits
Food and Preparations	Edible preparations, raw cooking materials and appliances
Clothing and Accessories	All kinds of fabrics, filament, clothing and accessories
Textiles	All kinds of textile articles and related products
Wood Products	Book, paper and other elementary products related to wood
Metals	Mining of all sorts of raw Metals
Refined and Processed Stones	Refined or processed stone products such as pearl and ceramics
Energy Drilling and Mining	Mining industry for energy, minerals, stone materials, etc.
Daily Instruments	Basic daily instruments such as sport requisites, timing and measuring tools
Raw Manufacturing	Basic products made from plastic, rubber, leather, etc.
Chemical Manufacturing	Advanced chemical products and pharmaceutical products
Chemicals	Raw chemical materials and by-products
Machinery and Construction	Machinery and mechanical appliances, including electrical equipment
Transportation	Vehicles and locomotives including aircraft, ship, railway, etc.
Weapons	Explosives, arms and ammunition
Services and Utilities	Financial, business, legal, communication, travel Services and Utilities, etc.
High-complexity Products	Agglomeration of advanced products with highest PCIs

Table 1. Categorization of Products

We further observe a positive correlation between the GDP and the export of High-complexity Products. A higher adjusted R-square exists between GDP and High-complexity Products exports, indicating that more successful economics, as measured by GDP, tend to be those that are capable of producing and exporting High-complexity Products as we define them. Tables 2a and 2b show the OLS results in a universe of 107 and 21 typical countries respectively.

	High-complexity Products	Food and Preparations	Animals and By-products
Adj. R-squared	0.075	0.006	0.002
F-statistic	9.556	1.638	1.175
No. Observations	107	107	107

Table 2a. OLS Regression Results of GDP against Percentage Volume of Exports

	High-complexity Products	Food and Preparations	Animals and By-products

Adj. R-squared	0.149	0.003	-0.012
F-statistic	4.510	1.059	0.7542
No. Observations	21	21	21

Table 2b. OLS Regression Results of GDP against Volume of Percentage Exports

3.2 Realized Industry Development

Based on the aforementioned product categorization, we aim to capture Singapore's realized industry production strategy between 2019 and 2024 on a complexity level. In Hausmann and Hidalgo (2009), distance is defined to be a measure of how far a country is from a certain product, not in its product space. We extend this concept to all product groups and quantify how technically equipped a country is to develop the corresponding product groups. In other words, the shorter the distance between a country and a product group, the more capable a country is of developing products within that group. Defining this net distance change as the Realized Industry Development (RID), our export data allow us to calculate and compare the RID of Singapore and the 18 product groups in 2024 and 2022 through the difference, where a positive RID indicates a shortening of the distance (i.e., advancement of the industry), and a negative RID indicates the reverse. It is notable that the number of industries with positive RID experienced a increase from 0 to 18 from 2022 to 2024.

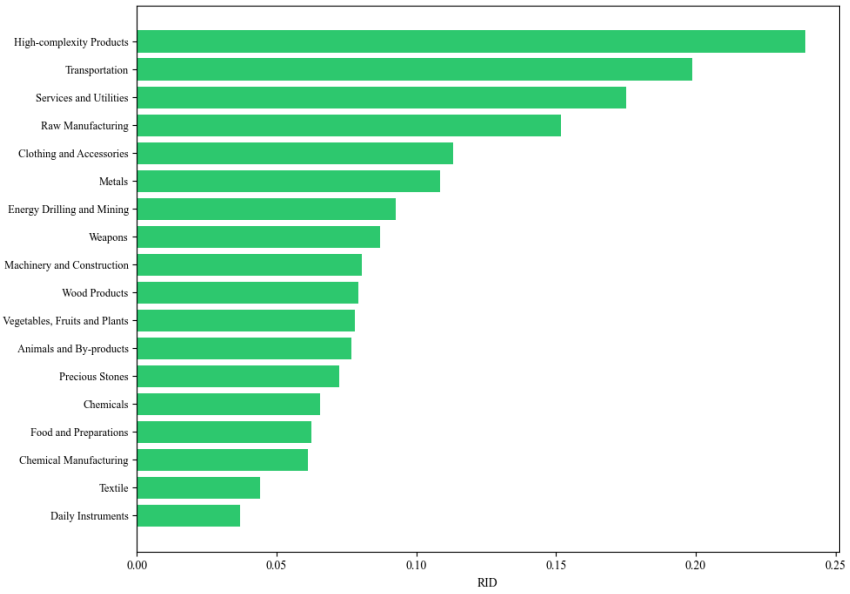


Table 3. Realized Industry Development (RID) of Singapore

3.3 Regional and Global Comparison

Table 4a below summarizes how the production strategy adopted by Singapore compares with selected leading economies in the world across different years, as measured in RID. Leading economies such as France, Greece, and Portugal gave a considerable level of attention to more advanced product groups such as Precious Stones and Chemicals in 2024 and Wood Products, Chemical Manufacturing, and Energy Drilling and Mining in 2022.

Rank	France	Greece	Portugal
#1	Precious Stones	Chemicals	Precious Stones
#2	Wood Products	Weapons	Chemical Manufacturing
#3	Vegetables, Fruits and Plants	Chemical Manufacturing	Wood Products
#4	Food and Preparations	High-complexity Products	Energy Drilling and Mining
#5	Energy Drilling and Mining	Clothing and Accessories	Food and Preparations

Table 4a. RID Comparison across Leading Economies (2024)

Table 4b further examines Singapore's development strategy against those of emerging economies on a global scale. Reasonably, emerging economies such as Malaysia, Thailand, and Indonesia have devoted the majority of their efforts to intermediate industry groups, showing moderate shifts in strategies over the course of the analysis period.

Rank	Malaysia	Thailand	Indonesia	Vietnam
#1	Chemical Manufacturing	Vegetables, Fruits and Plants	Chemical Manufacturing	Chemical Manufacturing
#2	High-complexity Products	Chemical Manufacturing	Machinery and Construction	Services and Utilities
#3	Machinery and Construction	Precious Stones	Textile	High-complexity Products
#4	Energy Drilling and Mining	Animals and By-products	Daily Instruments	Transportation
#5	Chemicals	Chemicals	Chemicals	Machinery and Construction

Table 4b. RID Comparison across Emerging Economies (2024)

4. Product Space Design

4.1 Suggested Development Vector

As outlined above, distance gives us an idea of how far each new product is from a country's current mix of exports. As defined in Hausmann and Hidalgo (2009), distance is "the weighted proportion of products connected to goods p that country c is not exporting, i.e., MCP = 0." Based on this definition, the distance between a country and any product that the country is currently exporting (i.e., the country is a significant exporter of the product as compared to the global average) is zero. Similarly, opportunity gain quantifies the contribution of a new product in terms of opening up the doors to more and more complex products. Intuitively, a product that a country is currently exporting (i.e., MCP = 1) has an opportunity gain of zero. An unreasonable investment in a certain product p would lead to a relative decrement in the export of more profitable products and therefore lead to a negative opportunity gain of product p. The opportunity gain is not the only estimator of a country's level of economic development. However, among countries with comparable economic strength, those having higher ECI, or in other words, more reasonable economic structures are often in better shape.

Singapore	South Korea	Japan	Germany	Netherlands
Chemicals	Chemicals	Chemicals	Transportation	Precious Stones
Precious Stones	Raw Manufacturing	Raw Manufacturing	Wood Products	Textile
High-complexity Products	Metals	Metals	Metals	Wood Products
Transportation	High-complexity Products	High-complexity Products	High-complexity Products	Services and Utilities
Daily Instruments	Transportation	Transportation	Daily Instruments	Clothing and Accessories
Services and Utilities	Daily Instruments	Weapons	Services and Utilities	Animals and By-products
Textile	Services and Utilities	Daily Instruments	Machinery and Construction	Vegetables, Fruits and Plants
Machinery and Construction	Textile	Services and Utilities	Chemicals	Food and Preparations
Clothing and Accessories	Machinery and Construction	Energy Drilling and Mining	Weapons	Energy Drilling and Mining
Weapons	Energy Drilling and Mining	Precious Stones	Clothing and Accessories	Chemicals
Energy Drilling and Mining	Clothing and Accessories	Textile	Chemical Manufacturing	Chemical Manufacturing
Chemical Manufacturing	Precious Stones	Wood Products	Textile	Raw Manufacturing
Wood Products	Weapons	Vegetables, Fruits and Plants	Raw Manufacturing	Weapons
Metals	Wood Products	Animals and By-products	Energy Drilling and Mining	Daily Instruments
Vegetables, Fruits and Plants	Chemical Manufacturing	Clothing and Accessories	Precious Stones	Machinery and Construction
Animals and By-products	Vegetables, Fruits and Plants	Machinery and Construction	Animals and By-products	Metals
Food and Preparations	Animals and By-products	Food and Preparations	Food and Preparations	Transportation
Raw Manufacturing	Food and Preparations	Chemical Manufacturing	Vegetables, Fruits and Plants	High-complexity Products

Table 5. Opportunity Gain Ranking on a Comparable Scale (2024)

We define the ranking of opportunity gain for a country c, as the Suggested Development Vector (SDV) of c. Table 5 summarizes how the SDV of Singapore compares with a selected group of comparable and leading economies in the world, with the industry groups ranked in decreasing order of significance.

Dark blue indicates positive opportunity gains, light blue indicates zero opportunity gains, and white indicates negative opportunity gains. The result shows that leading countries barely have any industries with positive opportunity gain. In other words, those countries are already in the most stable and efficient economic structure. Without the necessity of economic transition, those countries merely need to maintain a steady and balanced economic development. In contrast, developing countries have nearly half of industries with positive opportunity gain and share common interests in Chemicals and Precious Stones industries. It can be concluded that Singapore shares the characteristics of developing countries, with a balanced economic structure that leans towards intermediate-level industries.

South Korea	Japan	Netherlands	Israel
Chemicals	Chemicals	Precious Stones	Energy Drilling and Mining
Raw Manufacturing	Raw Manufacturing	Textile	Food and Preparations
Metals	Metals	Wood Products	Animals and By-products
High-complexity Products	High-complexity Products	Services and Utilities	Vegetables, Fruits and Plants
Transportation	Transportation	Clothing and Accessories	Chemicals
Daily Instruments	Weapons	Animals and By-products	Chemical Manufacturing
Services and Utilities	Daily Instruments	Vegetables, Fruits and Plants	Raw Manufacturing
Textile	Services and Utilities	Food and Preparations	Textile
Machinery and Construction	Energy Drilling and Mining	Energy Drilling and Mining	Precious Stones
Energy Drilling and Mining	Precious Stones	Chemicals	Transportation
Clothing and Accessories	Textile	Chemical Manufacturing	Daily Instruments
Precious Stones	Wood Products	Raw Manufacturing	Services and Utilities
Weapons	Vegetables, Fruits and Plants	Weapons	Wood Products
Wood Products	Animals and By-products	Daily Instruments	Machinery and Construction
Chemical Manufacturing	Clothing and Accessories	Machinery and Construction	Clothing and Accessories
Vegetables, Fruits and Plants	Machinery and Construction	Metals	Weapons
Animals and By-products	Food and Preparations	Transportation	Metals
Food and Preparations	Chemical Manufacturing	High-complexity Products	High-complexity Products

Table 6. SDV of Other Sample Countries (2024)

4.2 Budgeted Industry Development Strategy and Interpretation

The optimal budgeted industry development strategy aims to solve the most efficient allocations under the budgeted constraint of different situations of heuristic total export growth. In order to obtain such an allocation portfolio, we formulate an optimization problem as the following:

$$\underset{x \in \mathbf{R}^{|P|}}{\text{Max}} \quad \left(\frac{k(x) - < k(x) >}{stdev(k(x))}\right)_C$$

$$\text{Subject to}$$

$$\sum_{p \in P} x_p \leq B$$

$$x \geqslant 0$$

where k(x) is the second largest eigenvector of $M_{CC'}$ after adding x to the original export vector, C is the country that we are interested in, P is the product space, B is the heuristic budget.

If we can solve this problem, then we can know what the largest ECI to which a country can improve. Accordingly, the optimal solution x * that we obtain here is exactly our optimal allocation. However, the biggest problem is that the objective function is not convex. Therefore, we cannot apply the classical tools for convex optimization to this problem. In order to solve this problem, our idea is to use a greedy algorithm. The way that we apply the greedy algorithm is that firstly we separate the total budget into k equal parts, and then for the first equal part we try to search for the product that can optimize our objective function by

iterating through each $p \in P$. After we find the optimal product p for this equal part, we add the amount back to the export value of this product p and update the ECI for this country. If we observe that ECI does not change after iterating all the products, then we can add up one more equal part to invest. Until we find there is an ECI increment, we add the amount now to that product. The equal rest parts can be done in the same way.

By definition, a greedy algorithm is a kind of local search algorithm that is possible to run into a local minimum. Therefore, for local search algorithms, one typical way to keep it away from the local minimum is to enhance the vision of the algorithm by looking more steps forward. In our setting, whenever we obtain an increment of ECI, we mark down the required amount of export growth. Simultaneously, we continue searching for the next ECI increment point, compare the ratios of these two possibilities and pick the largest one. The ratio is defined as the following:

$$Ratio(p) = \frac{ECI_p - ECI_0}{||x_p - x_0||_1}$$

where ECI_p is just the ECI of the country after adding some amount of export increment to product p, which is exactly the amount that is able to make ECI change. Also, we can express it in the way of $||x_p - x_0||_1$.

In this way, we can obtain a further sighted greedy algorithm. Also, the number of parts that we would like to separate the budget into also matters. In order to solve this problem, we added a stabilizer on top of the enhanced greedy algorithm by running the algorithm for different numbers of parts and selecting the portfolio with the largest ECI increment and most complex economic structure.

Besides, it is quite common that there is a surplus in the budget. To deal with the surplus, we just add it to the product which needs the least amount of dollars to make $M_{cp} = 1$ (i.e., the smallest amount of required export increase).

4.3 Economic Panel

As stated above, the enhanced greedy algorithm is developed mainly for solving the optimal allocation strategy. In order to visualize this allocation, an economic panel is created, consisting of a pie chart, a line chart and a table. First, a pie chart is used to show the percentage of each product in the optimal portfolio. The number located in the center of the circle denotes the heuristic growth of total export, which is also our budget. Each category in the chart has a determined percentage of the allocated budget. The lines below the pie charts show how the ECI changes during the searching process of the greedy algorithm. Even though sometimes an increase of the export by 15% or 20% does not seem reasonable to developed countries, yet it can also give a sense of how a country should develop in the long term. Eventually, the graph in the end is just for numerical explanation for the pie charts.

4.5 Interpretation

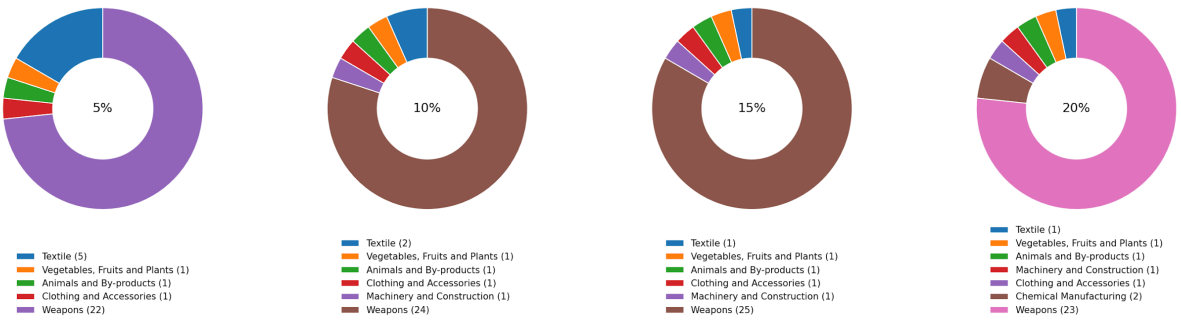
For the purpose of a clear explanation of the line charts, we label all the products in Table 7. Please refer to this table when trying to understand the investment trajectories in the line charts.

Label	Product Names
1	Animal and By-products
2	Vegetables, Fruits and Plants
3	Food and Preparations
4	Energy Drilling and Mining
5	Chemicals
6	Chemical Manufacturing
7	Raw Manufacturing
8	Wood Products
9	Clothing and Accessories
10	Textile
11	Metals
12	Machinery and Construction
13	Transportation
14	Weapons
15	Daily Instruments
16	Precious Stones
17	High-complexity Products
18	Services and Utilities

Table 7. Products Labeling

Figure 4 shows an economic panel for Singapore in 2024. ECI gains concentrate at the 5–10% budget levels and flatten beyond; the curve shows a step-change between 10% and 15% rather than smooth returns, indicating diminishing marginal complexity returns past moderate reallocation.

For 2024, a 5 percent increase in export would emphasize Weapons, followed by Weapons, Weapons, and Weapons for 10, 15, and 20 percent export growth respectively. These strategies go in line with the suggested trajectories based on ECI. Across budget levels, Weapons absorbs the majority of allocation, with Textile and Chemical Manufacturing entering only at the 20% threshold — the algorithm identifies a narrow set of high-marginal-complexity targets rather than broad reallocation.



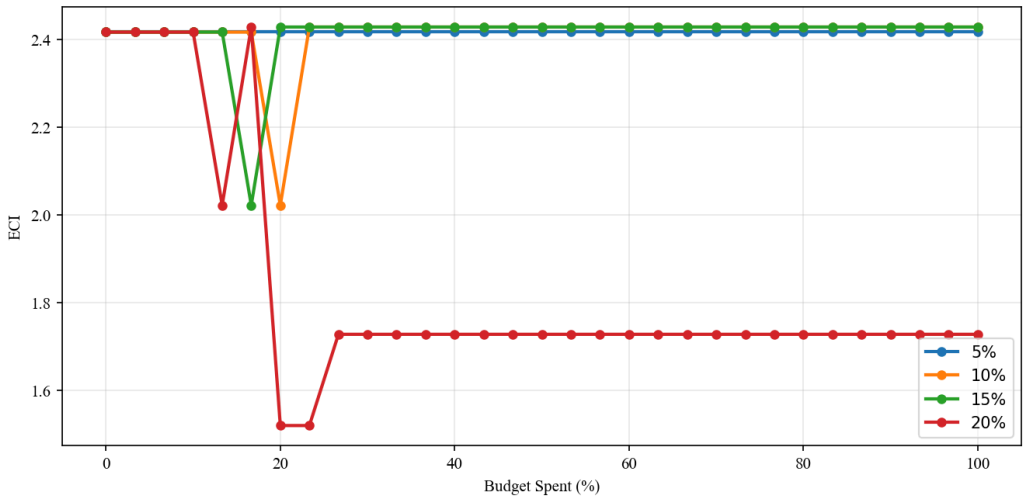


Figure 4. Greedy-Based ECI Improvement Results of Singapore with Different Budgets (2024)

	labels	5%	10%	15%	20%
0	Heuristic Export Growth	$27.1B	$54.3B	$81.4B	$108.5B
1	Animals and By-products	$0.90B	$1.81B	$2.71B	$3.62B
2	Vegetables, Fruits and Plants	$0.90B	$1.81B	$2.71B	$3.62B
3	Textile	$4.52B	$3.62B	$2.71B	$3.62B
4	Clothing and Accessories	$0.90B	$1.81B	$2.71B	$3.62B
5	Weapons	$19.90B	$43.41B	$67.83B	$83.21B
6	Machinery and Construction	0	$1.81B	$2.71B	$3.62B
7	Chemical Manufacturing	0	0	0	$7.24B
8	Increased ECI (Δ)	2.4166	2.4282	2.4282	1.7275

Table 8. Portfolio of Singapore (2024)

4.6 Sensitivity Analysis and Contingency Strategy

In the following section, our broad aim is to understand how sensitive Singapore is to changes in export in various industries. While the Economic Complexity Index has been shown to contain valuable information about the economic strength/capacity of a country, little has been said as to the stability of the measure itself. We seek to uncover the main drivers of stability under a country's ECI score and better understand longer-term projections for ECI growth. Additionally, gaining an intuition on the industries that are bolstering an economy and the industries that offer the greatest return per unit of investment is best done through visualization of industry-specific sensitivity. To this end, we have developed two categories of sensitivity analysis.

4.7 ECI Sensitivity Graph

First, in order to assess the effect particular industries have on each respective country, we developed ECI Sensitivity graphs (Figure 5). We use the shocking methodology to increment or decrement an industry's export level by a certain percentage and recalculate the ECI, all else being equal. While freezing the export dynamics of the rest of the world and enacting an instantaneous shock in export for a particular industry and country may not reflect real-life dynamics, what it does offer us is an understanding of a country's level of dependency on an industry in the context of ECI. The shocking is done in small percentage increments in order to smooth out the resulting curve. Additionally, a recurring theme one will see is that for most industries, a breaking point occurs where the ECI either jumps upwards or downwards, creating a hockey stick shape rather than a linear trend. While the inflection point may not be accurate in the context of real life, comparing the various inflection points across industries can give us valuable insight.

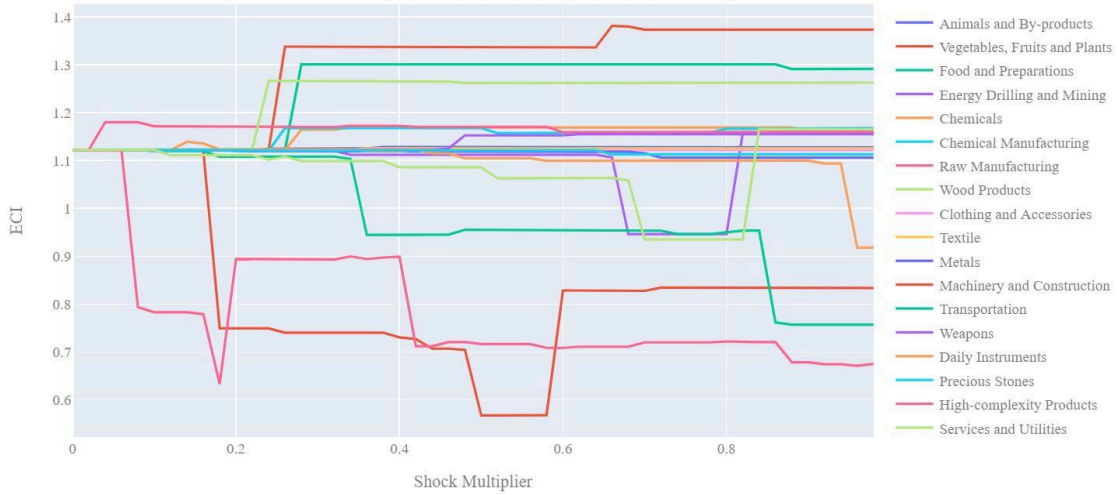


Figure 5. Sample ECI Sensitivity Graph

The graphs are formulated as follows. The first chart, Incremental ECI Sensitivity, shows positive shocking in which export numbers are shocked upwards per industry. The X-axis is in terms of percentage shock per current dollar value of export. The Y-axis is the recalculated ECI level post-shocking. The second chart, Decremental ECI Sensitivity, follows the same format except the X-axis represents negative shock. In other words, exports for each industry are decremented, and the ECI is accordingly recomputed.

4.8 Sensitivity Analysis Results and Interpretation

We examine Singapore's ECI Decremental and Incremental Sensitivity Charts for2024. An increase in Transportation will lead to the highest change in ECI,followed by High-complexity Products and Energy Drilling and Mining. Under positive shocks, ECI decreases with additional investment in Transportation, confirming it as the primary lever for complexity growth. For negative shocks, High-complexity Products emerges as the most

impactful industry, while reducing investment in Transportation causes ECI to move downward. The rest of the industries do not appear to have the same level of sensitivity.

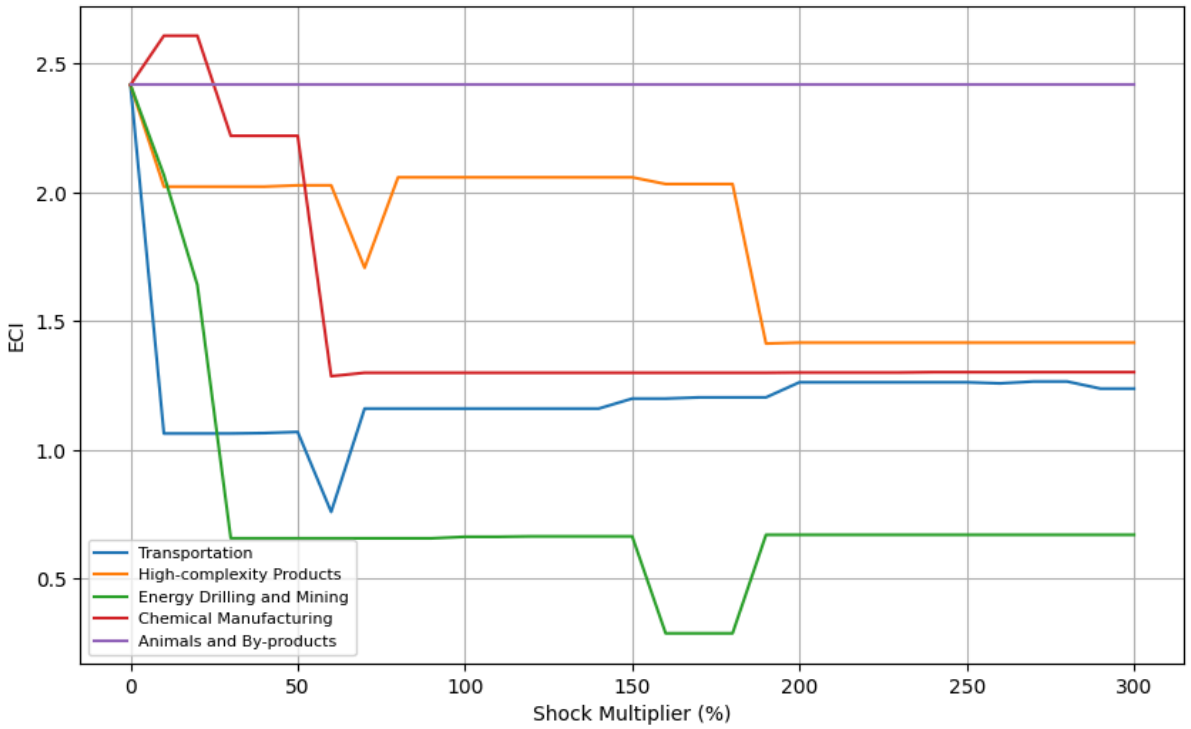


Figure 6a. Incremental ECI Sensitivity for Singapore (2024)

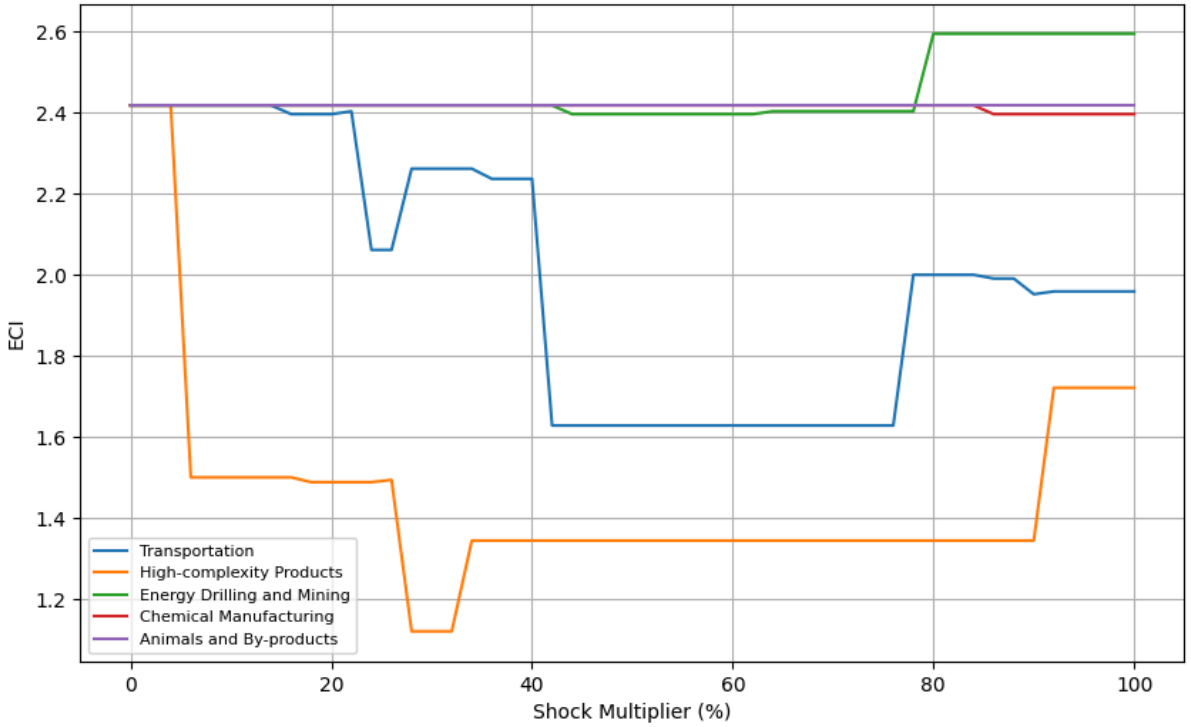


Figure 6b. Decremental ECI Sensitivity for Singapore (2024)

5. Macro-level Predictions

5.1 Structural Optimality Index

ECI sensitivity described in the section before has proven to be a significant factor in deciding the economic outcomes of countries. In order to quantitatively evaluate how effective our development strategies are and how deviations from the suggested developmental path will lead to suboptimal growth results, we define the measure – Structural Optimality Index (SOI) – on a country level as the Adjusted R-Square obtained from regressing the country's realized industry development over the suggested development strategy. Both strategies are quantified in nature, with the realized development represented by increasing export in every major industry group and the suggested development strategy represented by the opportunity gains of developing the corresponding industry groups as calculated in previous sections. The adjusted R-square from the regression thus provides us with a quantified measure of how the development path adopted by the country has fitted with/differed from calculations on a complexity level.

After obtaining this measure for all countries in our database, we further regress countries' five-year economic growth, which is defined as the percent increase in GDP (USD), over their SOI. Figure 9 illustrates the regression output, which shows a strong positive correlation between countries' SOI and their five-year GDP growth, achieving an adjusted R^2 of 0.0036%. In other words, the more a country develops, as suggested by calculations on a complexity level, the better economic outcomes it achieves.

Table 14: OLS Regression Results	
Dep. Variable: GDP	**F-Statistic:** 1.5051
R^2: 0.0109	**Prob (F-Stat):** 0.222
Adj R^2: 0.0036	**No. Observations:** 139

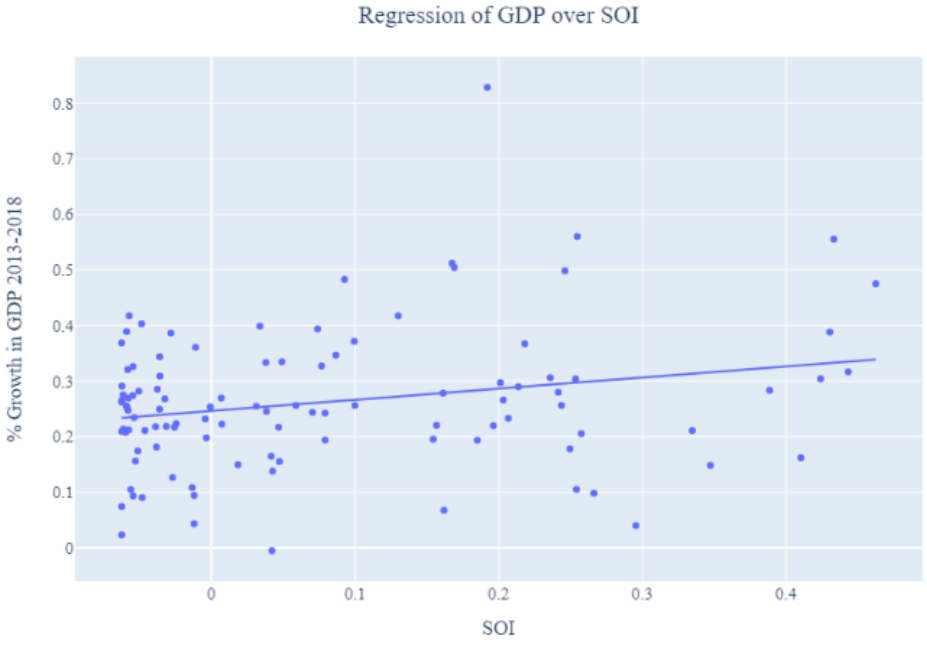


Figure 9. Regression of Five-Year GDP Growth over SOI (2019-2024)

5.2 SOI of Singapore and Its Implications

During the examined period, Singapore has a medium level of structural optimality as compared with all countries for which data is available. As the regression in Figure 10 suggests (the red curve represents export increase and the blue curve stands for opportunity gain of the corresponding industry), Singapore demonstrates a medium level of correlation with the suggested developmental path, having an adjusted R-square of around 0.0036.

Singapore ranks in the Q3 quartile among all countries in terms of SOI and recorded an average annual GDP growth rate of 7.0% from 2019 to 2024, placing it above its developed peers — consistent with the regression result in Figure 9. Vietnam, by contrast, sits in the top SOI quartile and recorded ~6.5% annual GDP growth, illustrating that higher SOI tracks with growth outcomes even at lower starting income levels.

Specifically, Singapore's high economic progress from 2019 to 2024 can be at least partially explained by a development strategy that demonstrates a medium level of alignment with our suggested developmental path. In particular, Singapore has moderately channeled resources toward Animals and By-products, Vegetables, Fruits and Plants, Food and Preparations, Energy Drilling and Mining, Chemical Manufacturing, Raw Manufacturing, Wood Products, Metals, Weapons — sectors where the complexity opportunity is more limited — while underinvesting in None, which offer greater complexity upside.

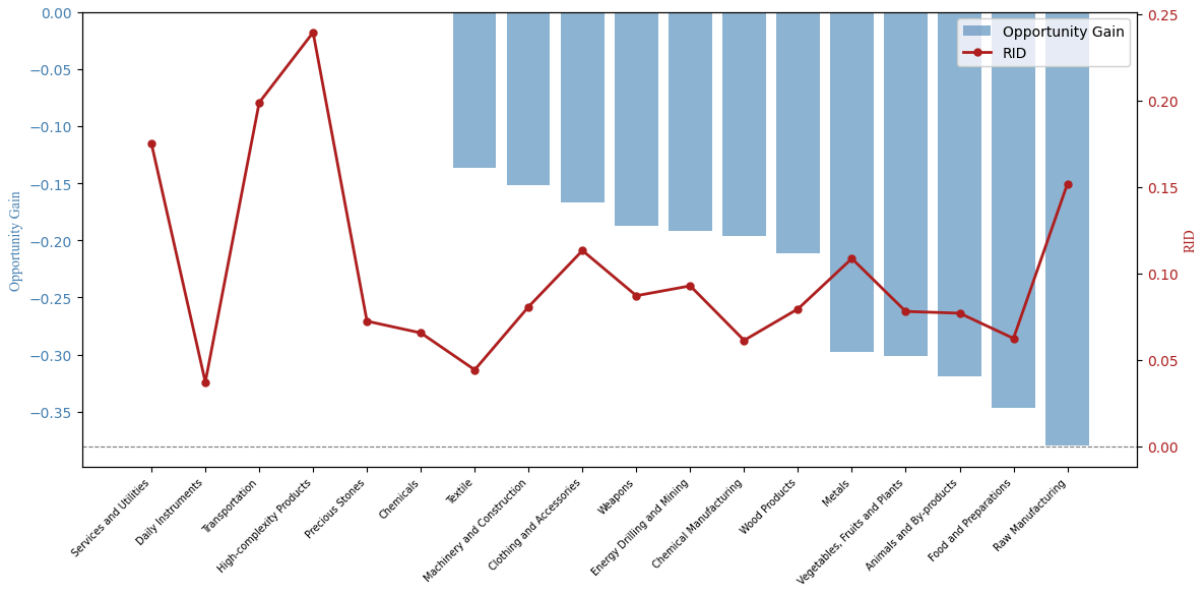


Figure 10. SOI of Singapore (2019-2024)

6. Concluding Remarks

In conclusion, Singapore has demonstrated a high level of economic growth in terms of annual GDP increase during the period from 2019 to 2024, concurrent with a significant shift in economic complexity as compared with its peer economies. An average annual GDP growth rate of 7.0% placed Singapore above its developed peers, and ECI increased by +3.692 over the same window.

We believe that Singapore's trajectory can be partially explained through the analysis of the economic and product complexity structure on the industry group level. The product classification system we adopt allows us to measure the realized industry development for various industry groups and to observe that Singapore progressed in developing the majority of its industries during the period, with the count of positively developing industries increasing from 0 to 18 between 2022 and 2024.

Based on our analysis of opportunity gain and sensitivity across major industry groups, Singapore can further its gains by prioritizing Weapons, Chemical Manufacturing, and Textile. Sensitivity analysis further identifies Transportation as the sector whose marginal expansion yields the largest positive shift in ECI. We further note that Singapore has moderately channeled resources toward Animals and By-products, Vegetables, Fruits and Plants, Food and Preparations, Energy Drilling and Mining, Chemical Manufacturing, Raw Manufacturing, Wood Products, Metals, Weapons while underinvesting in None, indicating a medium level of structural alignment with complexity-guided development, having an adjusted R-square of 0.0036. Since a positive correlation exists between five-year GDP growth and the structural optimality index as we define and calculate, Singapore's moderate structural alignment is consistent with its high growth record, and suggests that measurable gains remain available through a more complexity-guided development strategy.

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